

Overclocking the p-bit: Stochastic Binary Neural Network Inference with Autocorrelated sMTJ Randomness

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Abstract—Stochastic magnetic tunnel junctions (sMTJs) generate true random bits at roughly $10^6\times$ lower energy than software pseudorandom generators, making them attractive entropy sources for probabilistic-bit (p-bit) neural hardware. A standard design rule requires sampling an sMTJ no faster than about half its intrinsic fluctuation time, so that successive bits are uncorrelated; existing systems obey this rule by construction, and its actual cost to an application has not been measured. We treat the sampling ratio $r_0 = \Delta t/\tau_{\text{corr}}$ as a continuous design knob and measure, in simulation, how stochastic binary neural network (SBNN) classification accuracy on MNIST and Fashion-MNIST degrades as devices are queried up to $100\times$ faster than the rule allows. We find two sharply different regimes. When device states re-equilibrate between inputs, accuracy is almost unaffected: a single-sample classifier is provably immune to autocorrelation, and a 32-sample classifier loses under one accuracy point at $100\times$ overclocking. In continuous streaming operation, accuracy collapses below $r_0 \approx 0.5$ because stale device states leak information between consecutive inputs. Device-to-device variability does not mitigate this collapse; correlation-aware training largely does, recovering +47 points at $r_0 = 0.02$ for a 1–3 point cost at nominal speed, and extending the accuracy–throughput Pareto frontier several-fold beyond the 2τ -rule ceiling. Our results turn a conservative device-level constraint into a quantitative, workload-dependent design trade.

Index Terms—probabilistic computing, p-bit, stochastic magnetic tunnel junction, binarized neural networks, random telegraph noise, energy-efficient inference

I. INTRODUCTION

GENERATING randomness is surprisingly expensive on conventional hardware: algorithmic pseudorandom number generation at gigabit-per-second rates costs approximately 30 nJ per bit on CPUs and 3 nJ per bit on GPUs [1]. A stochastic magnetic tunnel junction (sMTJ)—a nanoscale magnet whose free layer flips between two resistance states under thermal agitation—produces physically random bits at roughly 0.03 pJ per bit [1], a gap of about six orders of magnitude. This gap motivates probabilistic-bit (p-bit) computing [2], [3], in which the sMTJ is not a peripheral random number generator but the stochastic neuron itself: an input current biases the device so that its time-averaged state follows $\langle m \rangle = \tanh(I)$ [2], exactly the activation required by stochastic binary neural networks (SBNNs) and Boltzmann machines.

Every physical entropy source, however, has a clock. An sMTJ flips on its own schedule, set by its Néel–Arrhenius fluctuation time; if it is read faster than it flips, successive

reads return copies of the same state rather than fresh coin flips. Daniels *et al.* showed that the autocorrelation between adjacent samples vanishes once the sampling interval exceeds roughly twice the mean dwell time [4], and this “ 2τ rule” has since functioned as a hard design constraint: systems are clocked slowly enough to satisfy it [5], and randomness quality is certified by statistical test suites that presume i.i.d. sampling [1], [6]. For devices engineered to the nanosecond regime [7], [8] the rule is cheap to obey; for the many demonstrated devices with microsecond-to-millisecond dwell times [9], it dictates the system clock.

What the rule has never been given is a price. No published work, to our knowledge, measures what actually happens to an application—rather than to a randomness test suite—when the rule is deliberately broken. This paper asks that question for classification: *how much faster than its correlation time can you sample an sMTJ neuron before a stochastic binary classifier fails, and what recovers the loss?*

We answer it in simulation with a physically grounded model: each hidden unit is a two-state telegraph process whose escape rates are biased by its synaptic input, reproducing the p-bit activation law exactly in its stationary distribution while exposing the sampling ratio $r_0 = \Delta t/\tau_{\text{corr}}$ as a continuous knob. Our contributions:

- **Two regimes, not one threshold.** If device states re-equilibrate between inputs (“reset” operation, natural for batch inference), classification is nearly immune to overclocking: single-sample inference is *exactly* unaffected for any r_0 (we give the argument in Sec. III), and 32-sample majority inference loses < 0.7 accuracy points at $r_0 = 0.02$, i.e. $100\times$ past the 2τ rule. In continuous streaming operation, by contrast, accuracy collapses below $r_0 \approx 0.5$, driven not by staleness within one input but by device state leaking across consecutive inputs.
- **Variability does not help.** A natural conjecture is that device-to-device dispersion in the energy barrier desynchronizes staleness across neurons and softens the collapse. We find the opposite: because dwell times are exponential in the barrier, dispersion produces a slow tail of extra-stale devices that dominates, and accuracy strictly degrades with barrier spread (by up to 5 points at $\sigma_\Delta = 1.5 k_B T$ in the collapse region).
- **Correlation-aware training recovers the streaming regime.** Training the network while its activations are sampled from the same correlated telegraph process flattens the failure curve: at $r_0 = 0.02$ streaming, accuracy

risks from 41% to 88% at a cost of 1–3 points at nominal speed. With physically-accounted throughput (reset operation pays a $\sim 2\tau$ re-equilibration wait per input), this extends the accuracy–throughput Pareto frontier several-fold beyond the 2τ -rule ceiling.

The result reframes autocorrelation from a defect to be engineered away into a budget to be spent: batch-style workloads can already run far past the nominal rule on existing devices, while streaming workloads define a genuine boundary that training, not fabrication, moves.

II. BACKGROUND AND RELATED WORK

A. p-bits and sMTJ entropy sources

A p-bit is a binary stochastic unit whose output at each query is $m = \text{sgn}\{\text{rand}(-1, 1) + \tanh I\}$, so that $\langle m \rangle = \tanh(I)$ [2]. Superparamagnetic MTJs realize this physics natively: a low energy barrier ($E_b \sim 1.4 k_B T$ for representative easy-plane devices [1]) lets the free layer fluctuate thermally, with dwell times following the Néel–Arrhenius law $\tau = \tau_0 \exp(\Delta/k_B T)$, $\tau_0 \approx 10 \text{ ps} - 1 \text{ ns}$ [2]. Demonstrated fluctuation timescales span six orders of magnitude: 2–5 ns autocorrelation times in easy-plane devices [7], nanosecond random telegraph noise in in-plane junctions [8], and microsecond-to-millisecond dwell times in the perpendicular devices used in the first p-bit factorization demonstration [9]. System-level demonstrations sample conservatively; e.g., a recent CMOS-plus-nanomagnet prober oversamples its $\sim 20 \text{ ms}$ relaxation time by $40\times$ [5].

B. Randomness quality and neural inference

Prior robustness studies of MTJ-based neural hardware concentrate on *static* imperfections: process-variation-induced shifts of the activation sigmoid [10], [11], and fabricated-array parameter spread with compensation via retraining-free weight correction [12]. On the randomness side, sMTJ generators are validated against NIST statistical suites [1], [6], and a recent study of software PRNGs found a threshold effect of generator quality on MNIST training [13]. Experimentally measured sMTJ bitstreams have been used for stochastic-computing inference [14], but with whatever correlation the apparatus happened to have—correlation was not a controlled variable. The *temporal* axis—autocorrelation as a swept, parameterized quantity mapped to task accuracy—is to our knowledge unexplored; existing work treats the 2τ decorrelation condition [4] as a constraint to satisfy rather than a dimension to characterize. That noise structure can have an optimal, nonzero operating point in neural systems has precedent in computational neuroscience [15], and tolerance of device imperfection is a recurring theme in compute-in-memory accelerator design [16]; we bring both perspectives to the p-bit sampling question.

III. MODEL

A. The sMTJ neuron as a biased telegraph process

Each hidden unit is a two-state fluctuator $s \in \{-1, +1\}$ with input-modulated Néel–Arrhenius escape rates

$$k_{-\rightarrow+} = f e^{+I}, \quad k_{+\rightarrow-} = f e^{-I}, \quad (1)$$

where I is the (dimensionless) synaptic input and $f = f_0 e^{-\delta\Delta}$ is the device attempt rate, with $\delta\Delta$ a per-device energy barrier offset in units of $k_B T$. The stationary distribution is

$$P(s=+1) = \frac{k_{-\rightarrow+}}{k_{-\rightarrow+} + k_{+\rightarrow-}} = \frac{1 + \tanh I}{2} \equiv p, \quad (2)$$

recovering the p-bit activation law exactly [2]: the *mean* behavior of every neuron is untouched by our model at every operating point, and only the *time structure* of its samples changes. The correlation time follows from the total rate,

$$\tau_{\text{corr}}(I) = \frac{1}{k_{-\rightarrow+} + k_{+\rightarrow-}} = \frac{1}{2f \cosh I}, \quad (3)$$

with the physically sensible property that strongly driven (saturated) neurons decorrelate faster than undecided ones.

Queried at interval Δt , the process obeys the exact two-state Markov propagator

$$P(s_{t+\Delta t}=+1 \mid s_t) = p + (\mathbb{K}[s_t=+1] - p) e^{-\Delta t/\tau_{\text{corr}}}, \quad (4)$$

which we use directly—there is no time-discretization error. The single control knob is the *sampling ratio*

$$r_0 = 2f_0 \Delta t = \Delta t/\tau_{\text{corr}}(0)|_{\delta\Delta=0}, \quad (5)$$

the query interval in units of the nominal zero-bias correlation time, so that $\Delta t/\tau_{\text{corr}} = r_0 e^{-\delta\Delta} \cosh I$. The regime $r_0 \gtrsim 2$ satisfies the 2τ rule [4]; $r_0 \ll 1$ is deliberate overclocking. Barrier dispersion is modeled as $\delta\Delta \sim \mathcal{N}(0, \sigma_\Delta^2)$, fixed per device per chip, giving a log-normal spread of correlation times while leaving every stationary firing probability—and hence all time-averaged statistics—unchanged. This isolates the purely temporal effect of variability, complementary to the static sigmoid-distortion studies of [10]–[12].

Model idealizations: we neglect read-disturb (the query does not perturb the device), voltage-dependent asymmetries, $1/f$ resistance noise, and temperature drift; the two-state approximation is standard for superparamagnetic MTJs in the telegraph regime [7], [8].

B. Network, training, and operating modes

We use a multilayer perceptron 784–256–128–10 with stochastic bipolar hidden activations: each hidden unit computes pre-activation I and outputs a fresh sample $s = \pm 1$ with $P(+1)$ given by Eq. (2) *at every query, including inference*—the p-bit operating style, as opposed to binarized networks that are deterministic at test time [17], [18]. The real-valued input layer and readout layer follow standard binarized-network practice. Baseline networks are trained with ideal i.i.d. randomness using the straight-through estimator [19], passing the gradient of the mean activation $\tanh I$; correlation-aware networks are trained identically except that hidden activations are sampled from the telegraph model (Eq. (4)) with device state running continuously across training minibatches.

At inference we distinguish two physically meaningful operating modes. In **reset** mode, device states re-equilibrate between inputs (equivalently, the hardware idles a few τ_{corr} per input); the T stochastic passes used for one input remain mutually correlated through Eq. (4). In **streaming** mode the devices are never idle: state also persists from each input to the

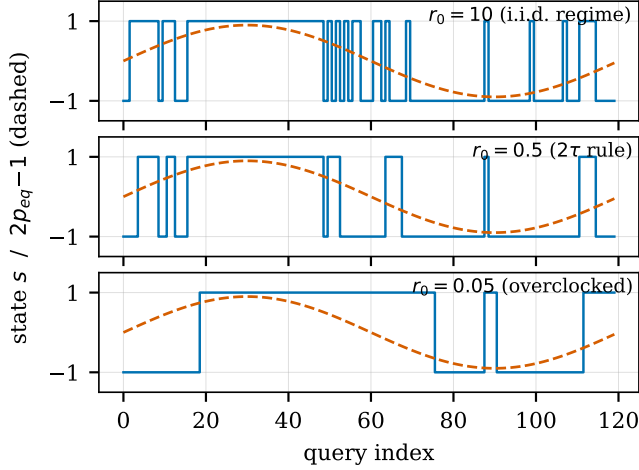


Fig. 1. Telegraph SMTJ neuron at three sampling ratios, driven by a slowly varying firing probability (dashed). At $r_0=10$ samples are effectively i.i.d.; at $r_0=0.05$ the device returns long runs of stale state.

next, modeling continuous pipelined operation. Class scores are averaged over the T passes.

Why reset-mode single-shot inference is exactly immune: For $T=1$ with equilibrated initial state, each neuron’s single sample is drawn from its stationary law, Eq. (2), regardless of r_0 : autocorrelation affects only the *joint* law of successive samples, not any single marginal, and with layers evaluated once per pass there is no second sample for it to correlate with. Overclocking therefore cannot change $T=1$ reset accuracy at all—a null that doubles as a correctness check on the simulator (Sec. V).

IV. EXPERIMENTAL SETUP

All experiments use MNIST [20] (60k/10k) with Fashion-MNIST [21] replication, full 10,000-image test sets throughout. Networks train for 30 epochs with Adam (10^{-3} , cosine decay), reaching 97.9% ($T=32$, ideal randomness) on MNIST and 88.6% on Fashion-MNIST. Sweeps cover $r_0 \in \{10, 5, 2, 1, 0.5, 0.2, 0.1, 0.05, 0.02\}$ and $T \in \{1, \dots, 64\}$, each point averaged over three independent noise seeds (and five chip seeds for variability experiments); error bars are standard deviations. Headline findings were replicated across five independently trained networks with agreement within ± 0.5 points. The simulator’s stationary statistics and autocorrelation decay were validated against Eqs. (2)–(4) by unit tests (marginal error $< 10^{-2}$; ACF decay-rate error < 0.02 at lags 1–4). Experiments run on CPU in hours; code, all per-run records, and exact configurations are available from the author.

V. RESULTS

A. Two regimes of failure (E1–E2)

Fig. 2 shows the central result. In reset mode, overclocking is almost free: at $r_0 = 0.02$ —100 $\times$ past the 2τ rule— $T=32$ accuracy drops only from 97.85% to 97.16%, and $T=1$ accuracy is bit-for-bit independent of r_0 (96.91% everywhere),

matching the exactness argument of Sec. III and validating the simulator. The mechanism is benign: within-input autocorrelation merely shrinks the effective number of independent samples toward the $T=1$ floor, and that floor is high.

Streaming is qualitatively different. Accuracy collapses below $r_0 \approx 0.5$, reaching 39.8% ($T=1$) and 97.4% $\rightarrow$ 41.0% ($T=32$) as $r_0 \rightarrow 0.02$. The damage is done *between* inputs: a stale device state carries the previous image’s activation pattern into the current one. Averaging helps—each $4\times$ increase in T buys roughly a $2\text{--}3\times$ overclocking margin—but cannot restore the i.i.d. anchor. Notably, even at near-frozen $r_0=0.02$ streaming accuracy stays well above the $\sim 10\%$ constant-output floor: by Eq. (3), strongly driven neurons decorrelate $\cosh I$ -fold faster, so the input “burns through” the stale state via its most saturated units. Fashion-MNIST reproduces the entire structure (reset: 88.6% $\rightarrow$ 87.3% at $r_0=0.02$; streaming: collapse to 40.8%), indicating the phenomenon is not dataset-specific.

B. Device variability does not desynchronize the damage (E3)

Since dwell time is exponential in the barrier, one might hope dispersion σ_Δ spreads correlation times so that at any moment some neurons are fresh—a stochastic-resonance-like rescue with precedent in balanced spiking networks [15]. Fig. 3 refutes this for our setting: at fixed r_0 , streaming accuracy is monotonically worse with increasing spread (78.5% $\rightarrow$ 73.5% at $r_0=0.05$ as $\sigma_\Delta : 0 \rightarrow 1.5 k_B T$), while reset mode is indifferent. The asymmetry explains it: a fast-tail device cannot beat “fresh”—there is no headroom above i.i.d.—while a slow-tail device (τ larger by $e^{+\delta\Delta}$) becomes catastrophically stale, so the log-normal slow tail dominates the network-level effect. For fabrication, this inverts the usual reading of barrier uniformity requirements [12]: uniformity matters *more*, not less, when a system will be overclocked.

C. Correlation-aware training (E4)

Training with the telegraph source active (state running continuously across minibatches) produces networks whose streaming robustness curves are nearly flat (Fig. 4). The network trained at $r_0=0.2$ holds 87.6% at $r_0=0.02$ where the ideal-trained baseline falls to 41.0%; the $r_0=0.5$ -trained network is the better choice for moderate overclocking (94.1% vs. 78.6% at $r_0=0.05$) at only 1.1 points below baseline at nominal speed. The trade is the familiar robustness–accuracy exchange of noisy training, here induced by a *physical* noise process rather than an artificial regularizer: the network learns activation patterns that remain decodable when a fraction of each layer’s units echo their previous state.

D. The throughput Pareto frontier (E5)

Throughput accounting exposes the design consequence. Reset operation must idle $\sim 2\tau_{\text{corr}}$ per input to re-equilibrate, so its throughput saturates at 0.5 images/ τ_{corr} no matter how aggressively the T passes are overclocked; streaming pays no idle time but suffers the collapse; and correlation-aware training moves the streaming curve (Fig. 5). At a 97% accuracy

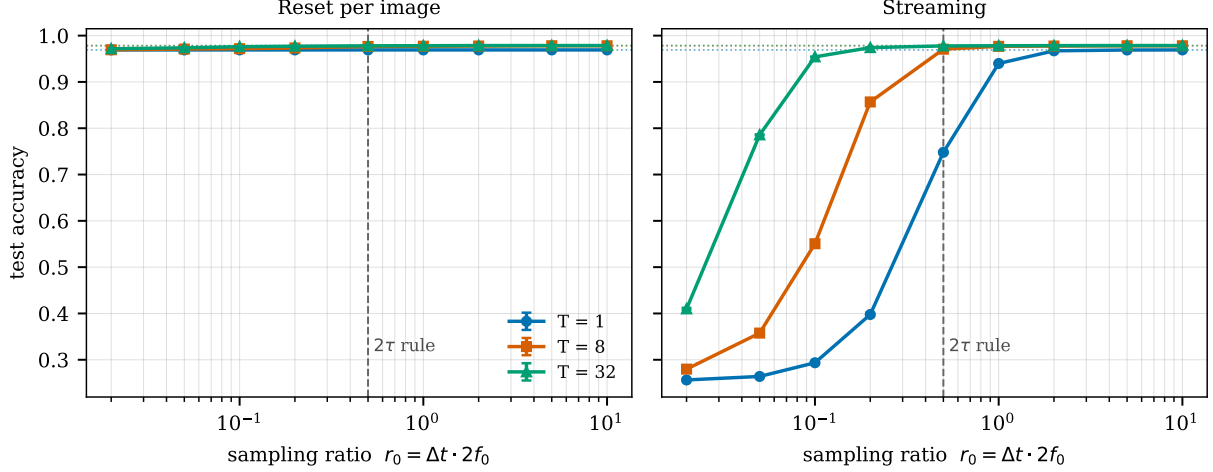


Fig. 2. Accuracy vs. sampling ratio r_0 (MNIST, mean $\pm$ std over 3 noise seeds; dotted lines: ideal-randomness anchors). **Left:** reset mode is nearly flat down to $r_0=0.02$; $T=1$ is exactly flat as predicted. **Right:** streaming mode collapses below $r_0 \approx 0.5$; larger T buys roughly a decade of margin.

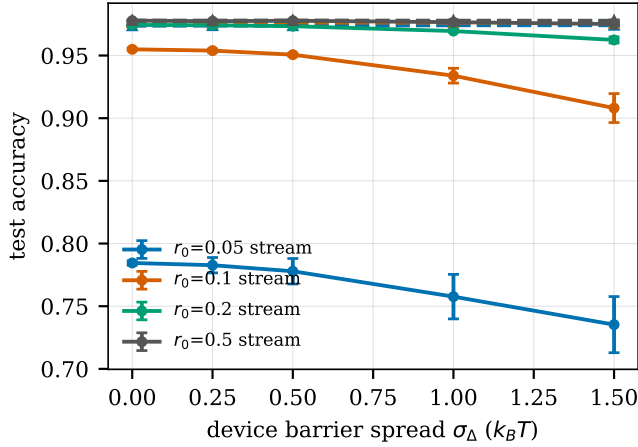


Fig. 3. Device-to-device barrier spread σ_Δ does not rescue streaming operation: accuracy strictly degrades with dispersion, most strongly deep in the overclocked regime (solid: streaming; dashed: reset; $T=32$, 5 chip seeds).

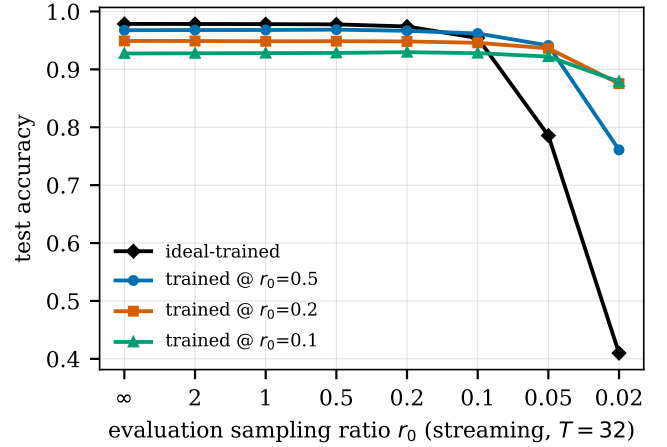


Fig. 4. Correlation-aware training flattens the streaming failure curve ($T=32$): a 1–3 point sacrifice at nominal speed removes the collapse, recovering +47 points at $r_0=0.02$.

target the frontier is reset operation with its averaging passes overclocked ($T=16$ at $r_0=0.02$: 97.1% at 0.43 images/ τ_{corr}), $1.7\times$ past the best rule-compliant operating point; at 95%, correlation-aware *single-shot* streaming at $r_0=1$ (95.9%, vs. 94.0% ideal-trained) doubles rule-compliant throughput; and at 90%, correlation-aware single-shot streaming at $r_0=0.5$ (93.4%) reaches 2 images/ τ_{corr} , a $4.0\times$ gain.

Concretely, for the millisecond-class devices of the first p-bit demonstrations [9], the 2τ rule limits a $\tau_{\text{corr}} \sim 10$ ms device to ~ 50 classifications/s at $\geq 95\%$ accuracy; the frontier in Fig. 5 makes 100 classifications/s at $\geq 95\%$, or 200/s at $\geq 90\%$, available from the same silicon. For nanosecond easy-plane devices [7] the same ratios apply to systems that are sampling-rate-limited rather than device-limited.

VI. DISCUSSION

Design guidance. The 2τ rule is the right specification for randomness-certification workloads (TRNG bitstreams, NIST

compliance [1], [6]), but for p-bit *inference* it is workload-dependent: (i) batch-style inference with per-input equilibration can overclock essentially without limit—the marginal-distribution argument guarantees it for $T=1$ and our sweeps bound the loss for $T \leq 64$; (ii) streaming inference obeys a genuine boundary near $r_0 \approx 0.5$, which fabrication (barrier uniformity) does not move but training does; (iii) if a system will be overclocked, device uniformity should be prioritized over device speed, since the slow tail of the dwell-time distribution, not the mean, sets the damage.

Limitations. This is a simulation study with an idealized two-state telegraph model; read-disturb, retention drift, and $1/f$ components in real devices [1] may tighten the boundaries reported here, and our networks are small MLPs on standard image benchmarks. The streaming results depend on input order only through the generic assumption that consecutive inputs are uncorrelated; correlated input streams (video) could

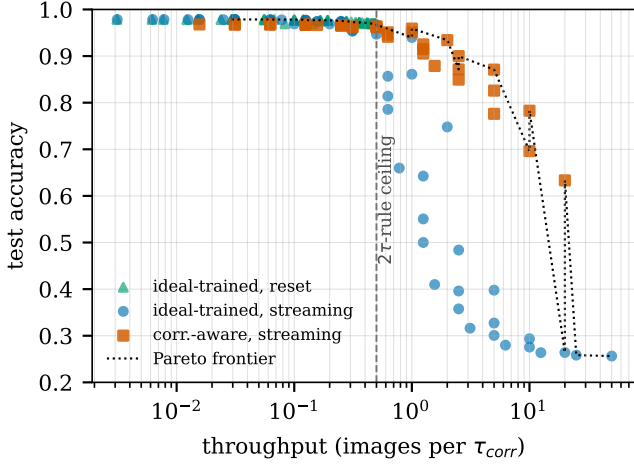


Fig. 5. Accuracy vs. physically-accounted throughput (images per τ_{corr}). Reset operation pays a $\sim 2\tau$ re-equilibration wait per input and cannot exceed 0.5 images/ τ_{corr} (vertical line); streaming with correlation-aware training extends the Pareto frontier beyond it.

interact with device staleness in either direction. Validating the r_0 sweep on fabricated arrays—e.g. the instrumented systems of [5]—is the natural next step, as is extending to convolutional SBNs and to temperature-induced bias drift, which we deliberately excluded as a separately characterized axis [1].

VII. CONCLUSION

We measured the application-level price of violating the sMTJ sampling rule and found it is not a wall but a fork: nearly free for equilibrated batch inference, a hard collapse for streaming inference—caused by cross-input state leakage, immune to device-variability remedies, and largely repaired by correlation-aware training. Treating the sampling ratio as a design variable rather than a constraint converts device autocorrelation into usable throughput: $2\times$ at matched 95% accuracy and $4\times$ at 90%, beyond the 2τ -rule ceiling, from training alone. We hope this reframing—and the accuracy-vs- r_0 curves as a device-to-application specification chart—is useful to both device and architecture communities.

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